

Awwab A. Azam

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Research interests: Theoretical and Computational Condensed-Matter Physics; Machine Learning

EXPERIENCE

Undergraduate Research Assistant

Apr 2026 - present

Bernevig Group, Department of Physics, Princeton University, Princeton, NJ

- Advisor: Prof. B. Andrei Bernevig
- Selected for the Princeton Center for Complex Materials (PCCM) REU, summer 2026.
- Performed density functional theory (DFT) calculations with Quantum ESPRESSO and EPW, including Wannierization, to compute superconducting transition temperatures T_c .
- Contributing to ongoing projects on diffusion models and machine learning force fields (MLFFs) for materials generation and property prediction.

Undergraduate Research Assistant

Oct 2024 - present

Yu Group, Department of Physics, University of Florida, Gainesville, FL

- Advisor: Prof. Jiabin Yu
- Designed, implemented and trained neural networks (NNs) with various architectures (including self-attention) to accelerate the calculation of reduced density matrices for quantum many-body systems ([arXiv:2511.07367](#)).
- Contributed to a follow-up study extending the method to fractional Chern insulators, with benchmarks across NN architectures ([arXiv:2605.20326](#)).
- Developed large-scale parallel exact diagonalization code using a matrix-free sparse Lanczos eigensolver with OpenMP. The code produced the numerical results in [arXiv:2509.05410](#).

Chief of Education

Apr 2025 - May 2026

IEEE Signal Processing Society, University of Florida

- Developed [the content for a two-part workshop series](#) on machine learning for physics and engineering, focusing on training NNs using the PyTorch library and implementing transformer models from scratch.
- Taught the workshop series twice (in summer and fall) in collaboration with the Society of Physics Students. Lessons included both live coding and theoretical explanations.

Peer Instructor (Teaching Assistant)

Aug 2025 - Dec 2025

Department of Electrical & Computer Engineering, University of Florida

- Peer Instructor for EEL 3701C (Digital Logic and Computer Systems) for the Fall 2025 semester.
- Responsibilities included leading lab sections, evaluating hardware/software demos, holding office hours, grading (labs/homework/quizzes/exams), and answering students' questions.

EDUCATION

2023-2027 (expected) BS (Computer Engineering) at **UF**

(GPA: 3.89/4.0)

- Selected Courses: Solid-State Electronic Devices (EEE 3396), Thermodynamics 1 (EML 3100), Signals and Systems (EEL 3135), Microprocessor Applications 1 (EEL 4744C), Digital Logic and Computer Systems (EEL 3701C), Data Structures and Algorithms (COP 3530), Computer Organization (CDA 3101)

2023-2027 (expected) BA (Mathematics) at UF

(GPA: 3.89/4.0)

– **Minor in Physics**

- Selected Courses (Physics): Statistical Physics (PHY 4523, *in progress*), Quantum Mechanics (PHY 4604), Electromagnetism (PHY 3323), Modern Physics (PHY 3101), Theoretical Physics (PHZ 3113), Enriched Physics 1/2 with Calculus (PHY 2060/2061)
- Selected Courses (Mathematics): Differential Equations 2 (MAP 4305), Real Analysis (MAA 4211), Abstract Algebra 1 (MAS 4301), Linear Algebra (MAS 4105), Probability Theory (STA 4321), Honors Elementary Differential Equations (MAP 2302)

Honors and Awards:

- UF Center for Condensed Matter Sciences (CCMS) Undergraduate Fellowship (2025-2026)
- AI Scholars Program (2025-2026, 2026-2027)
- President’s Honor Roll (Summer 2025)
- Dean’s List (Summer 2025, Fall 2025, Spring 2026)
- 3rd place winner, GatorHack AI Hackathon, sponsored by NVIDIA and Verizon (2023)

PUBLICATIONS, POSTERS, AND PRESENTATIONS

Publications

- **A. A. Azam**, B. Lian, S. Ryu, and J. Yu, “Wilson-Loop-Ideal Bands and General Idealization”, *Phys. Rev. Lett.* (2026), [arXiv:2509.05410 \[cond-mat.mes-hall\]](#). *Accepted in PRL (in press)*.
- J. B. Hart, **A. A. Azam**, T. Li, Y. Li, Y. Bi, H. Pan, and J. Yu, “Representability-Aware Neural Networks for Reduced Density Matrices: Application to Fractional Chern Insulators” (2026), [arXiv:2605.20326 \[cond-mat.str-el\]](#). *Under review at PRX Intelligence*.
- **A. A. Azam**, L. Zhao, and J. Yu, “Reduced Density Matrices Through Machine Learning” (2025), [arXiv:2511.07367 \[cond-mat.str-el\]](#). *In preparation for journal submission*.

Posters and Presentations

- **A. A. Azam**, “Generator Matching for Materials Modelling”, Princeton Center for Complex Materials (PCCM) REU Final Poster Session, Princeton University, Princeton, NJ, 29 July 2026. *Expository, based on Holderrieth et al.*, [arXiv:2410.20587](#).
- **A. A. Azam**, “Generator Matching for Materials Modelling”, PCCM REU Flash Talks, Princeton University, Princeton, NJ, 08 July 2026. *Expository, based on Holderrieth et al.*, [arXiv:2410.20587](#).
- **A. A. Azam**, L. Zhao, and J. Yu, “Machine-Learning Accelerated Calculations of Correlation Functions”, [APS Global Physics Summit 2026](#), Session Y30: Quantum Hall Effect & Computation of Topological Properties, Denver, CO, 20 Mar. 2026. *Contributed talk*.
- **A. A. Azam**, L. Zhao, and J. Yu, “Reduced Density Matrices through Machine Learning”, The Pennsylvania State University Condensed Matter Theory Group Meeting (online), 05 Dec. 2025. *Talk delivered by L. Zhao; co-prepared presentation and participated in Q&A*.
- **A. A. Azam**, L. Zhao, and J. Yu, “Machine-Learning Accelerated Calculations of Reduced Density Matrices”, Quantum Geometry and Moiré Materials Workshop (sponsored by the Institute for Fundamental Theory) Poster Session, University of Florida, 10 Nov. 2025.
- **A. A. Azam**, L. Zhao, Y. Ge, and J. Yu, “Machine Learning Acceleration for Hartree-Fock Calculations”, UF Undergraduate Fall Research Symposium, 16 Oct. 2025.
- **A. A. Azam**, “Introduction to PyTorch”, Workshop taught for the IEEE Signal Processing Society at UF (in collaboration with the Society of Physics Students) twice, on 15 May and 04 Sept. 2025.

- **A. A. Azam**, “Self-Attention and Transformer Models from Scratch”, Workshop taught for the IEEE Signal Processing Society at UF (in collaboration with the Society of Physics Students) twice, on 12 June and 20 Nov. 2025.

SKILLS

Programming Languages	Python, C/C++, Java/Kotlin
Machine Learning	PyTorch and JAX for training NN models, PyTorch Geometric for graph neural networks, hyperparameter optimization with RayTune and Optuna
Software	Quantum ESPRESSO and EPW for DFT and electron-phonon calculations, scientific programming using NumPy/SciPy, plotting and data visualization with Matplotlib, Bash scripting, Linux command line, high-performance computing (SLURM, OpenMP, HiPerGator 4.0/AI HPC cluster at UF), Git/GitHub, L ^A T _E X
Hardware	ARM/AVR assembly, microcontrollers (embedded C, UART/SPI), breadboard prototyping, basic FPGA programming with VHDL